

Saeed Biabani Mahalli

World Models • Large Language Models • Representation Learning • Multimodal Learning

Juybar, Mazandaran, Iran

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Research Profile

Undergraduate Computer Engineering student with research interests in world models, representation learning, multimodal learning, and large language models. Experienced in developing Transformer-based architectures for language and vision tasks, including OCR, retrieval-augmented generation, and generative modeling, with a focus on generalizable and efficient learning systems.

Research Interests

World Models & Model-Based Learning • Representation Learning • Self-Supervised Learning • Multimodal Learning • Vision-Language Models • Large Language Models • Efficient Transformer Architectures

Education

- **Bachelor of Science in Computer Engineering** | [University of Science and Technology](#)
Sep 2022 - Present, Mazandaran, Behshahr

Publications

- Biabani Mahalli, S., Pirgazi, J. & Ghanbari Sorkhi, A. *An effective knowledge-distilled transformer for intent detection and slot filling in Persian language*. J Supercomput 82, 369 (2026). <https://doi.org/10.1007/s11227-026-08432-6>
- Khajevand, M., Biabani Mahalli, S., Pirgazi, J. & Ghanbari Sorkhi, A. *VLT-PTR: Vision-LSTM and Transformer-Based Approach for Persian Text-Image Recognition*. Under Review

Experience

- **AI Research Engineer / AI Team Lead** | [mayanext.com](#)
Apr. 2025 - Present, Mazandaran, Behshahr

Notable Projects

- **Optical Character Recognition** | [Github](#)
Developed a Transformer-based model for Persian scene text recognition.
- **Intelligent Persian Drug Assistant** | [mayanext.com](#)
Developed a GraphRAG-based Persian medical assistant using Neo4j and LangChain for knowledge-grounded pharmaceutical question answering.

Relevant Skills

- **Machine Learning:** Transformers, LLMs, VLMs, Representation Learning, Self-Supervised Learning, RAG
- **Framework & Tools:** PyTorch, Hugging Face Transformers
- **Programming & Tools:** Python, Git, Linux, Docker, FastAPI.
- **Language:** Persian (Native), English (Working Proficiency)